

Remember our House? Let's give it a name

<http://example.com/sw2016/ourHouse>



```
presentation-example.ttl — /Users/hoekstra/projects/semanticweb-rdf-...
presentation-example.ttl
1  @prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
2  @prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
3
4  <http://example.com/sw2016/ourHouse>
5
6  # This is *not* valid Turtle
7
```

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Define a namespace prefix, to make it a bit shorter

ex:ourHouse



```
presentation-example.ttl — /Users/hoekstra/projects/semanticweb-rdf-...
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2  @prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
3  @prefix ex: <http://example.com/sw2016/> .
4
5  ex:ourHouse
6
7  # This is *not* valid Turtle
8
```

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... and there's our Forest

ex:ourHouse

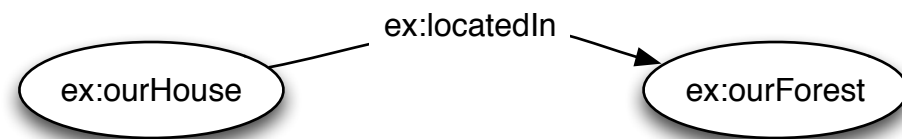
ex:ourForest



```
presentation-example.ttl — /Users/hoekstra/projects/semanticweb-rdf-...
presentation-example.ttl
1 @prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
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4
5 ex:ourHouse
6 ex:ourForest
7
8 # This is *not* valid Turtle
9
```

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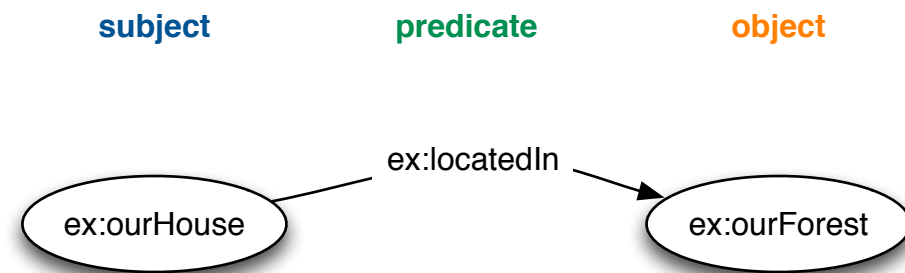
We create an ex:locatedIn relation between them



```
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presentation-example.ttl
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2 @prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
3 @prefix ex: <http://example.com/sw2016/> .
4
5 ex:ourHouse ex:locatedIn ex:ourForest .
6
```

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Remember? Subject, predicate, object

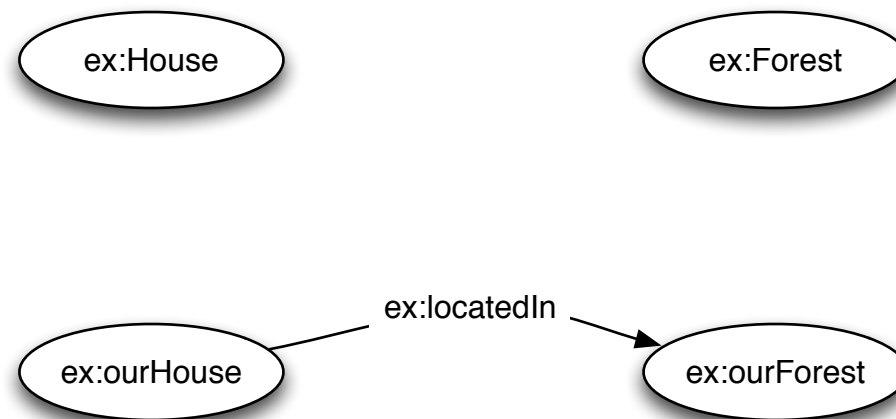


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3  @prefix ex: <http://example.com/sw2016/> .
4
5  ex:ourHouse ex:locatedIn ex:ourForest .
6
```

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Let's add two more generic terms

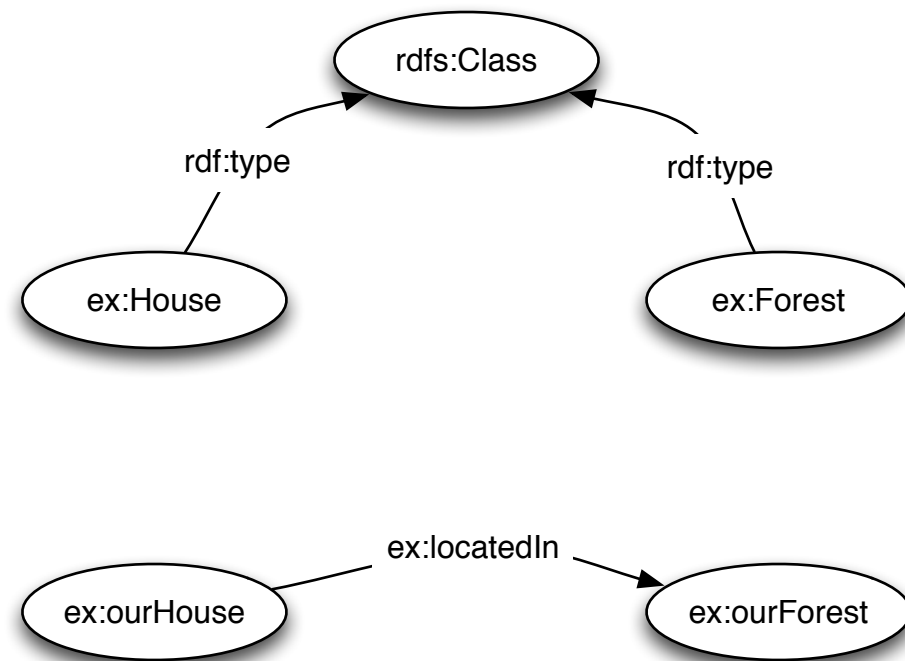


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presentation-example.ttl — /Users/hoekstra/projects/semanticweb-rdf-...

presentation-example.ttl
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3  @prefix ex: <http://example.com/sw2016/> .
4
5  ex:ourHouse ex:locatedIn ex:ourForest .
6  ex:House
7  ex:Forest
8
9  # This is *not* valid Turtle
10
```

File 0 Project 0 ✓ No Issues presentation-example LF UTF-8 Turtle 7 updates

Let's add to more generic terms, they will be our **classes**

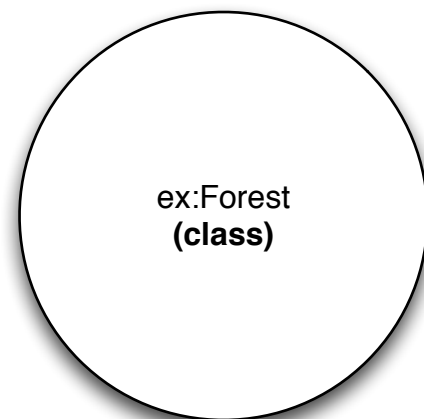
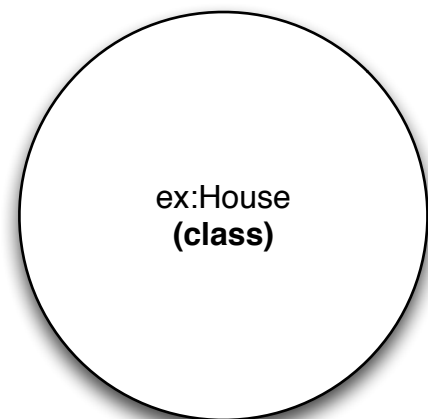
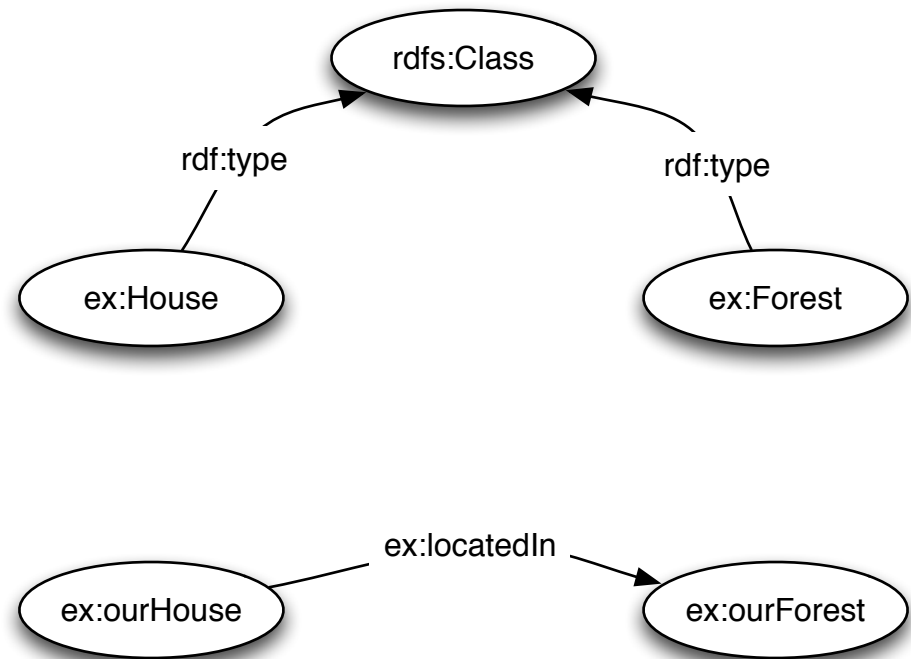


```
presentation-example.ttl — /Users/hoekstra/projects/semanticweb-rdf-...

presentation-example.t •
1  @prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
2  @prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
3  @prefix ex: <http://example.com/sw2016/> .
4
5  ex:ourHouse ex:locatedIn ex:ourForest .
6  ex:House   rdf:type     rdfs:Class .
7  ex:Forest  rdf:type     rdfs:Class .
8
9
10
```

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Classes are most easily understood as **sets** of instances (Venn diagrams)

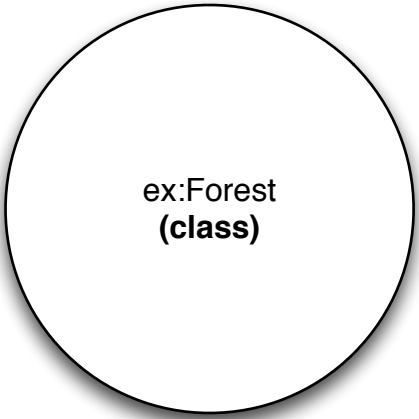
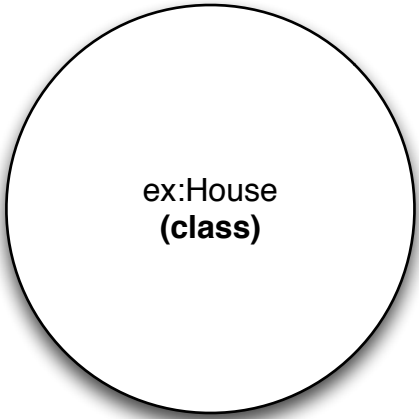
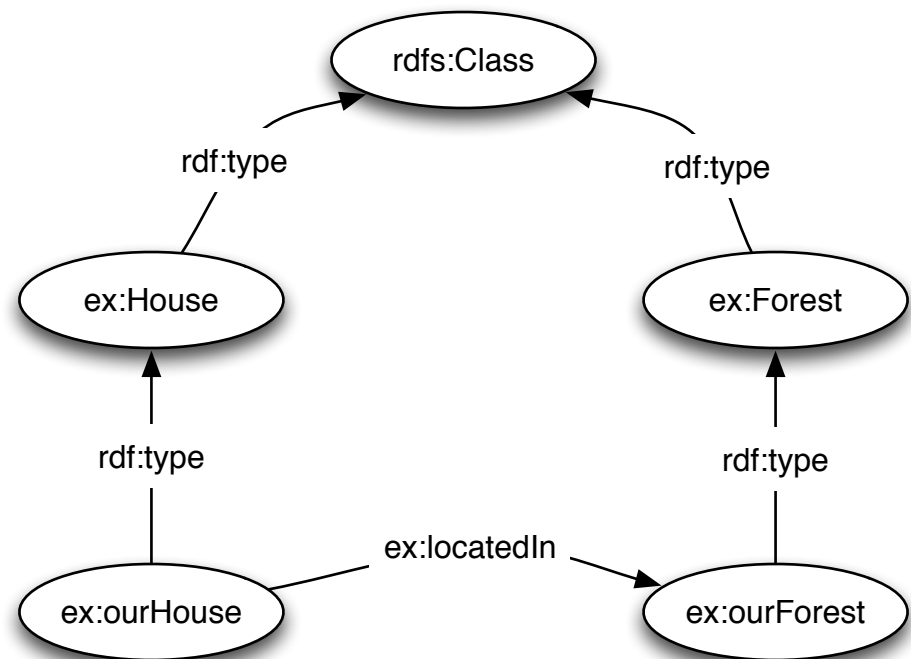


A screenshot of a code editor window titled `presentation-example.ttl`. The editor shows the following Turtle code:

```
1 @prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
2 @prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
3 @prefix ex: <http://example.com/sw2016/> .
4
5 ex:ourHouse ex:locatedIn ex:ourForest .
6
```

The editor's status bar at the bottom indicates: `File 0`, `Project 0`, `✓ No Issues`, `presentation-example`, `LF`, `UTF-8`, `Turtle`, and `7 updates`.

We can state that ex:ourHouse and ex:ourForest belong to the classes we just created

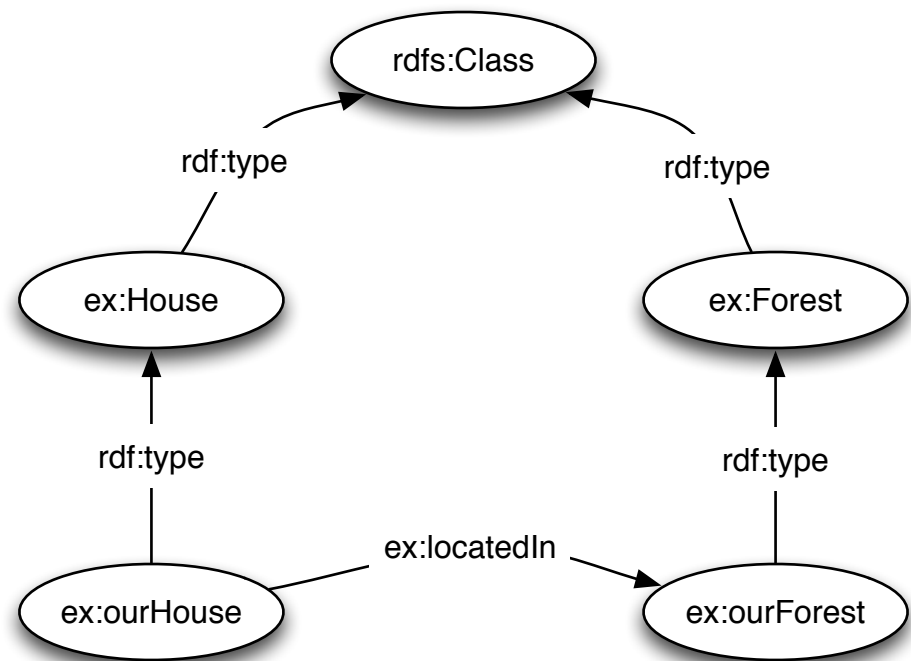


```
presentation-example.ttl — /Users/hoekstra/projects/semanticweb-rdf-...

presentation-example.ttl
1  @prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
2  @prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
3  @prefix ex: <http://example.com/sw2016/> .
4
5  ex:ourHouse ex:locatedIn ex:ourForest .
6  ex:House   rdf:type     rdfs:Class .
7  ex:Forest  rdf:type     rdfs:Class .
8
9  ex:ourHouse rdf:type     ex:House .
10 ex:ourForest rdf:type     ex:Forest .
11
12
13

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```

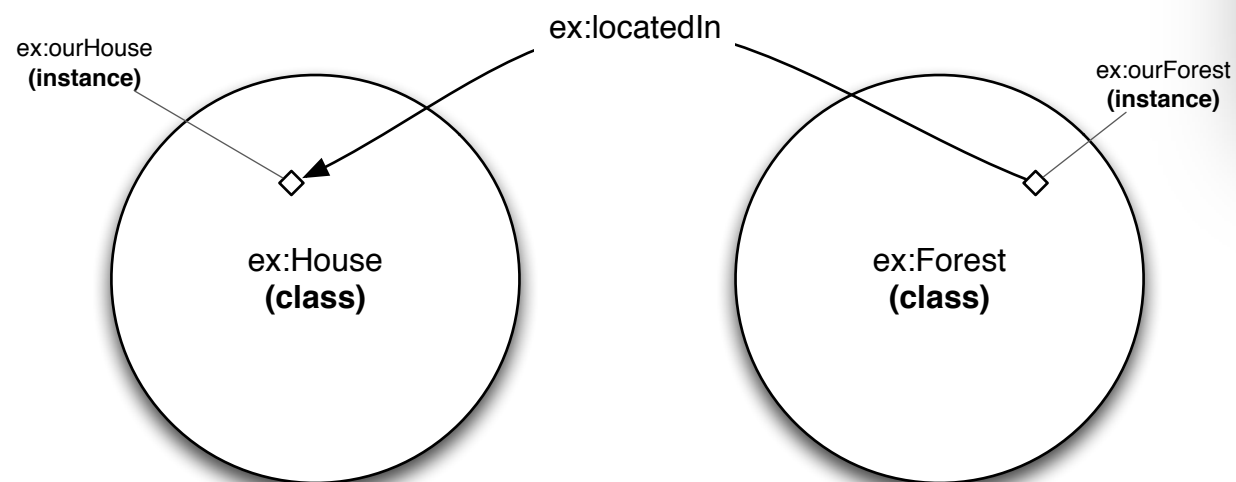
They become **members** or **instances** of the classes

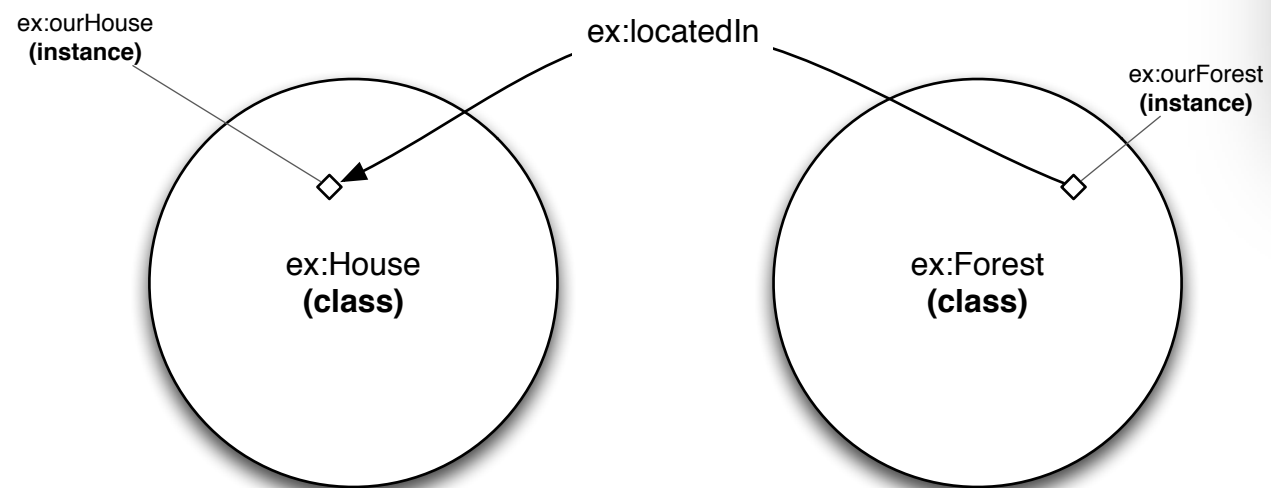
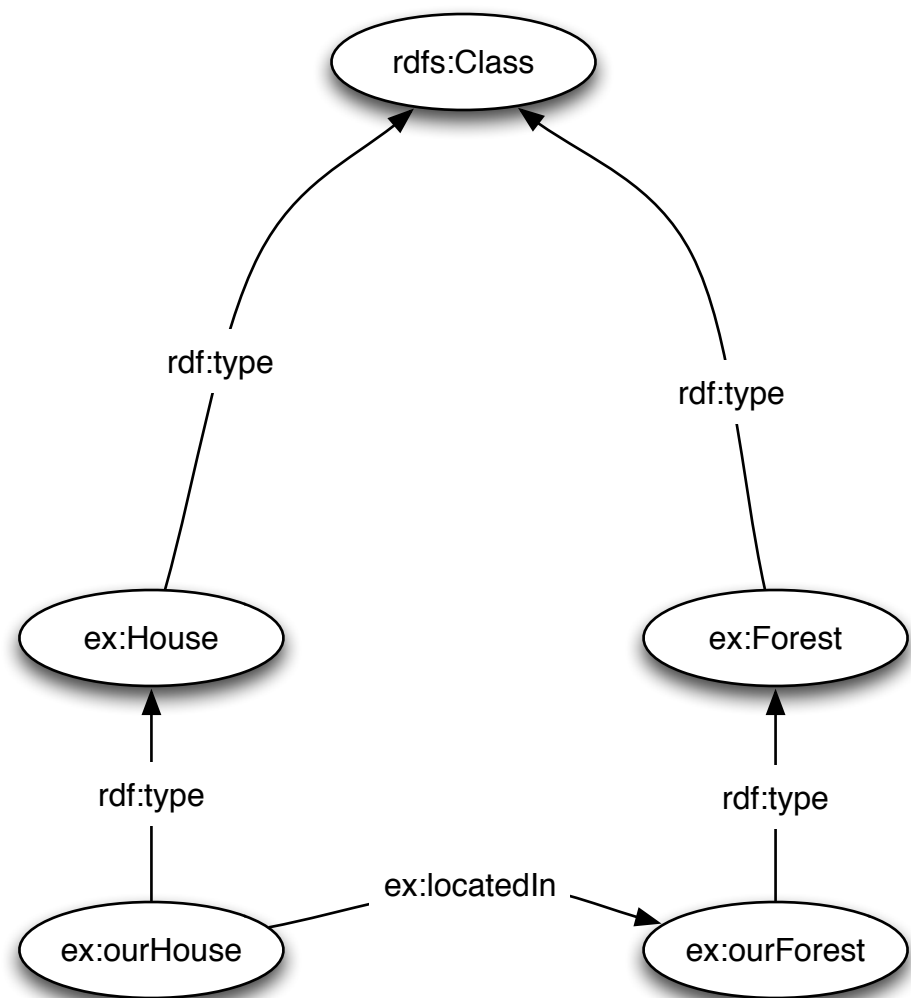


```
presentation-example.ttl — /Users/hoekstra/projects/semanticweb-rdf-...

presentation-example.t •
1  @prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
2  @prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
3  @prefix ex: <http://example.com/sw2016/> .
4
5  ex:ourHouse rdf:type ex:House ;
6              ex:locatedIn ex:ourForest .
7
8  ex:House rdf:type rdfs:Class .
9  ex:Forest rdf:type rdfs:Class .
10
11 ex:ourForest rdf:type ex:Forest .
12
13
14

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```



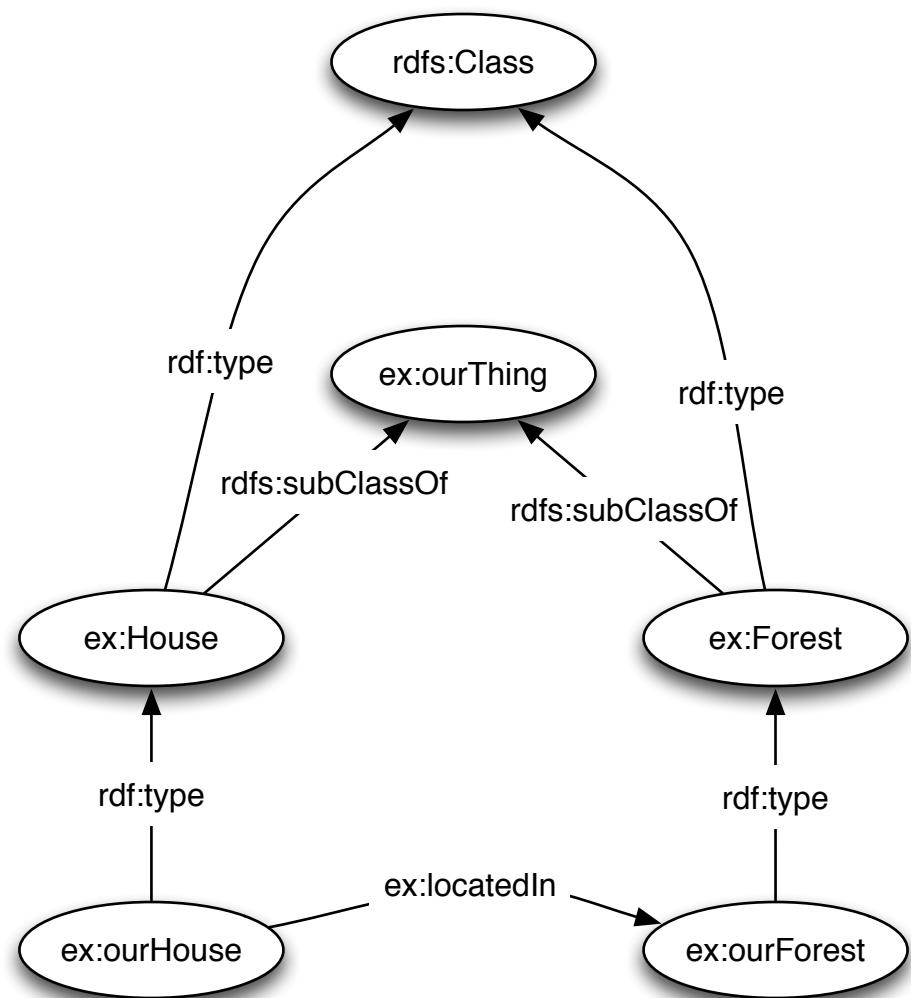


```
presentation-example.ttl — /Users/hoekstra/projects/semanticweb-rdf-...

presentation-example.ttl
1  @prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
2  @prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
3  @prefix ex: <http://example.com/sw2016/> .
4
5  ex:ourHouse rdf:type ex:House ;
6              ex:locatedIn ex:ourForest .
7
8  ex:House rdf:type rdfs:Class .
9  ex:Forest rdf:type rdfs:Class .
10
11 ex:ourForest rdf:type ex:Forest .
12
13
14
```

File 0 Project 0 ✓ No Issues presentation-example LF UTF-8 Turtle 7 updates

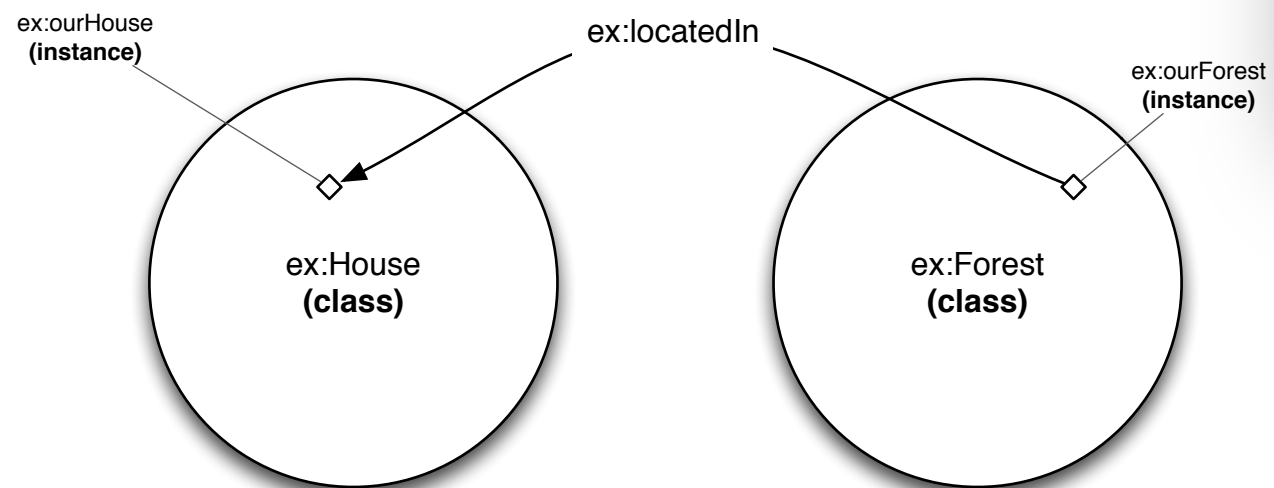
Let's make some room for more...



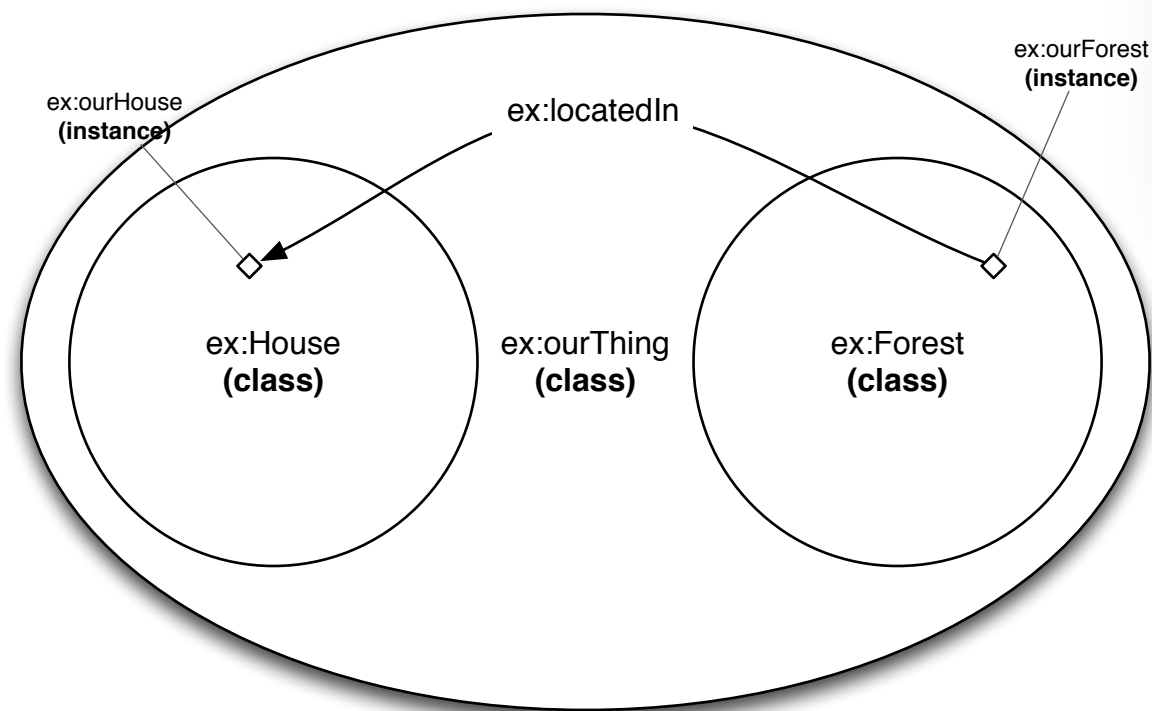
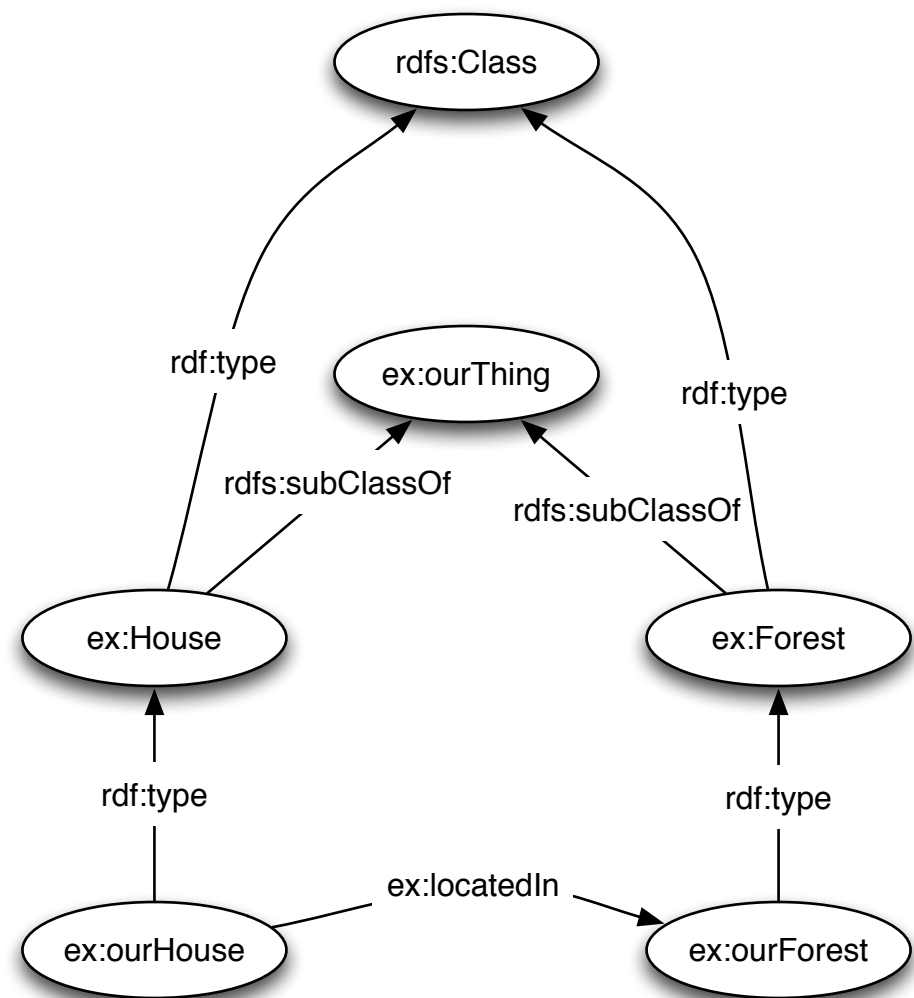
```
presentation-example.ttl — /Users/hoekstra/projects/semanticweb-rdf-...

presentation-example.ttl
1  @prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
2  @prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
3  @prefix ex: <http://example.com/sw2016/> .
4
5  ex:ourHouse rdf:type ex:House ;
6              ex:locatedIn ex:ourForest .
7
8  ex:House rdf:type rdfs:Class ;
9           rdfs:subClassOf ex:ourThing .
10
11 ex:Forest rdf:type rdfs:Class ;
12          rdfs:subClassOf ex:ourThing .
13
14 ex:ourForest rdf:type ex:Forest .
15
16
```

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We add a **superclass** "ex:ourThing" for our two classes

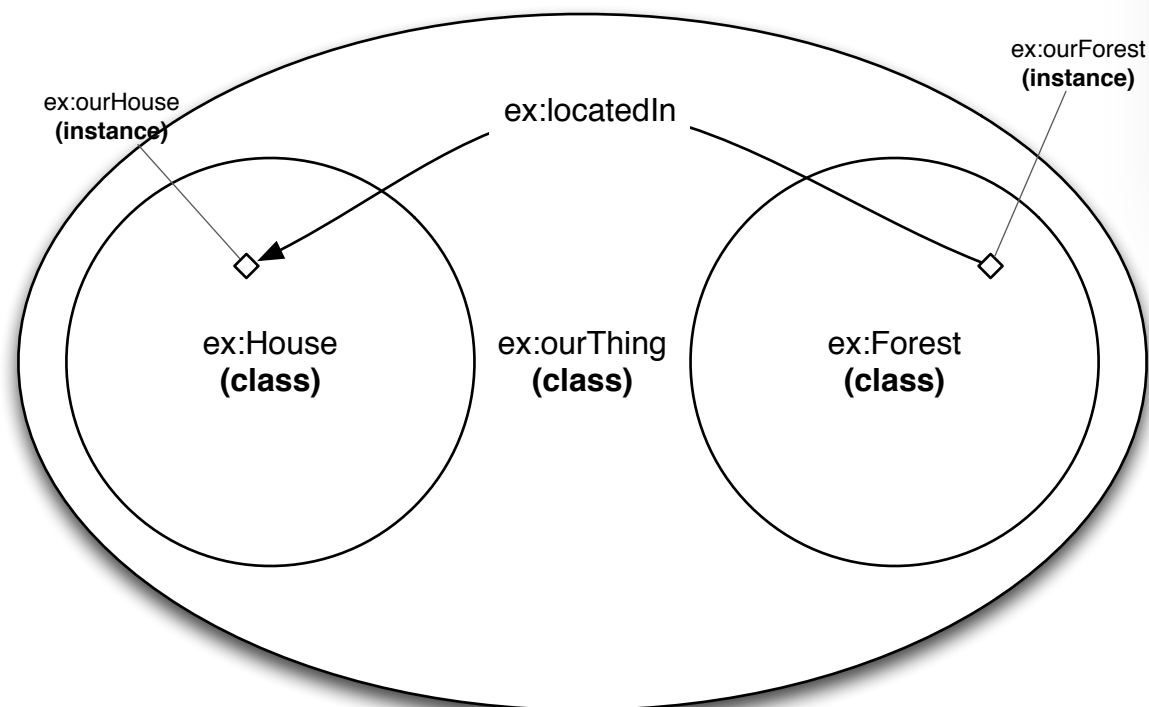
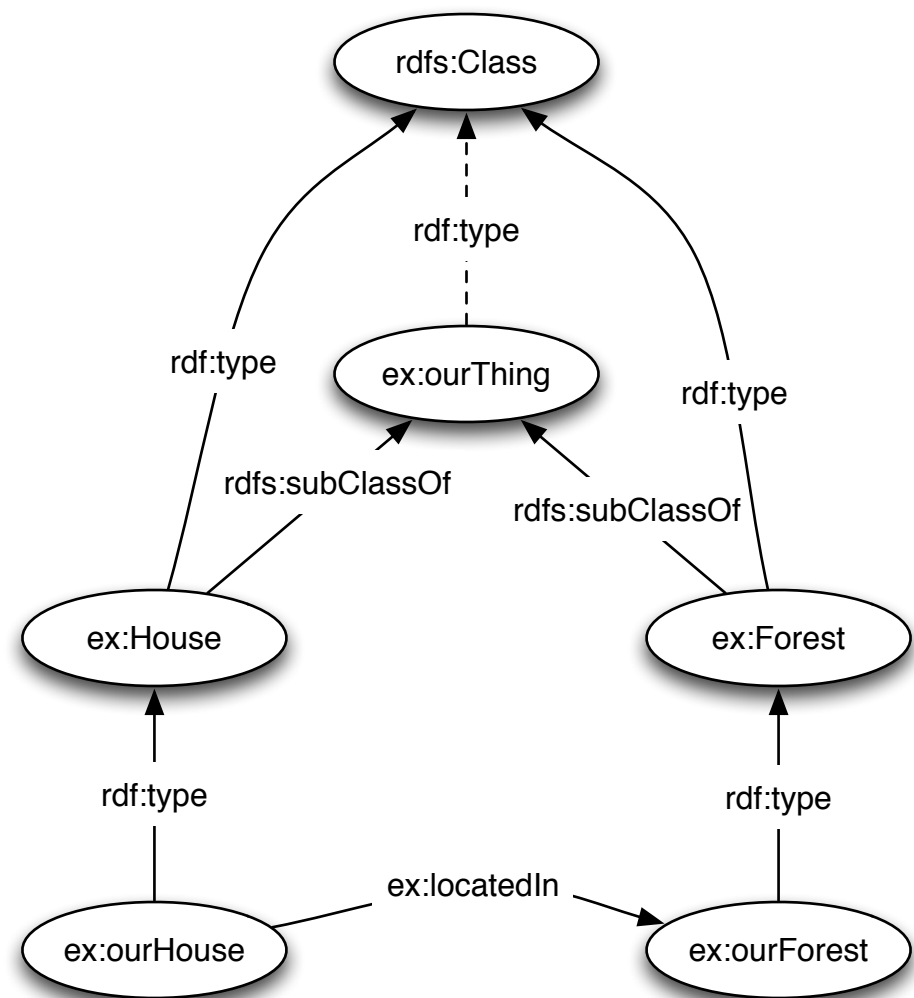


```
presentation-example.ttl — /Users/hoekstra/projects/semanticweb-rdf-...

presentation-example.ttl
1  @prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
2  @prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
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4
5  ex:ourHouse rdf:type ex:House ;
6              ex:locatedIn ex:ourForest .
7
8  ex:House rdf:type rdfs:Class ;
9           rdfs:subClassOf ex:ourThing .
10 ex:Forest rdf:type rdfs:Class ;
11          rdfs:subClassOf ex:ourThing .
12
13 ex:ourForest rdf:type ex:Forest .
14
15
16
```

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It encompasses every member of our two classes (and more)

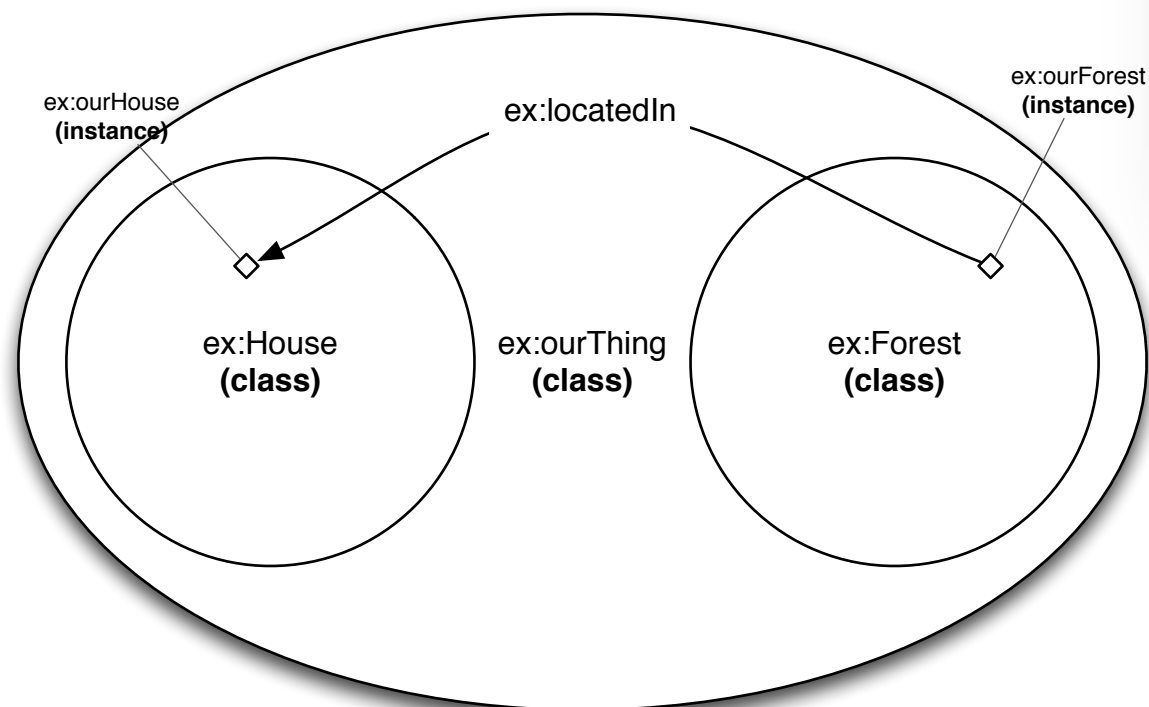
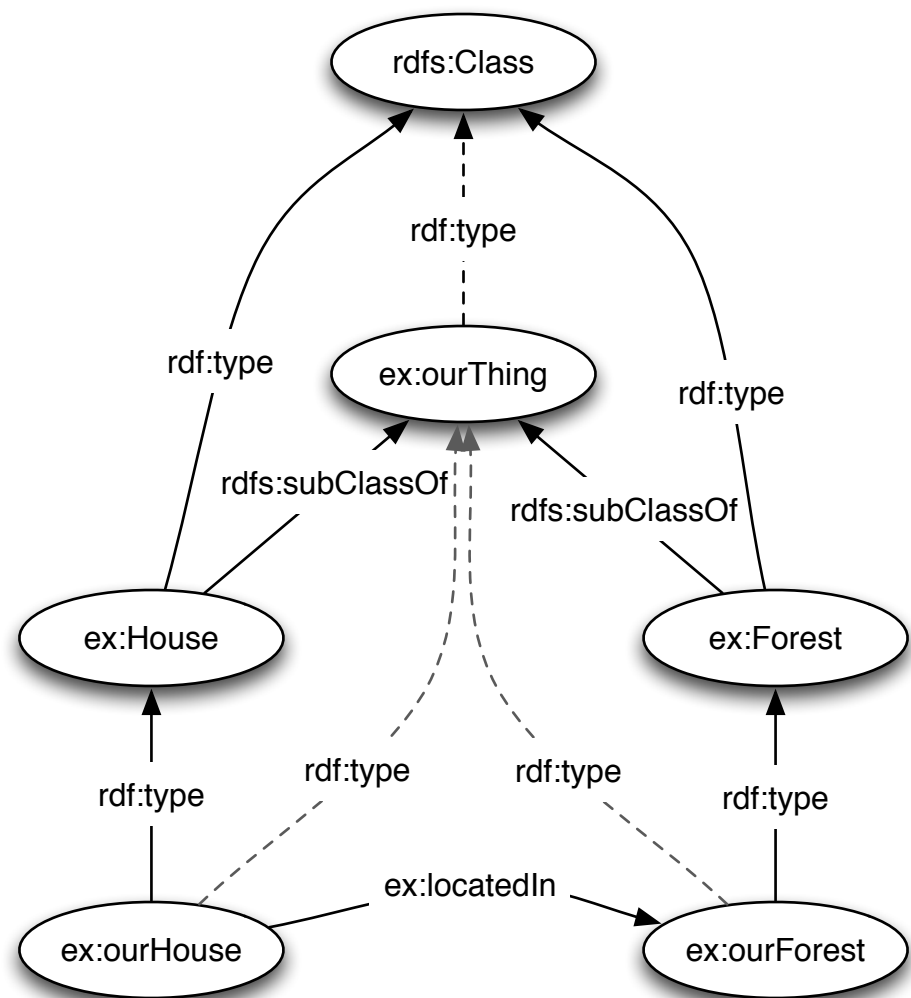


```
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presentation-example.ttl
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3  @prefix ex: <http://example.com/sw2016/> .
4
5  ex:ourHouse rdf:type ex:House ;
6              ex:locatedIn ex:ourForest .
7
8  ex:House rdf:type rdfs:Class ;
9           rdfs:subClassOf ex:ourThing .
10 ex:Forest rdf:type rdfs:Class ;
11          rdfs:subClassOf ex:ourThing .
12
13 ex:ourForest rdf:type ex:Forest .
14
15 ex:ourThing rdf:type rdfs:Class .
16
```

5.
$$\begin{array}{ccccc} s & & \text{rdfs:subClassOf} & & y \\ y & & \text{rdfs:subClassOf} & & z \end{array} \implies s \quad \text{rdfs:subClassOf} \quad z$$
6.
$$\begin{array}{ccccc} p & & \text{rdfs:subPropertyOf} & & q \\ q & & \text{rdfs:subPropertyOf} & & r \end{array} \implies p \quad \text{rdfs:subPropertyOf} \quad r$$
7.
$$\begin{array}{ccccc} s & & p & & o \\ p & & \text{rdfs:subPropertyOf} & & q \end{array} \implies s \quad & & q & & o$$
8.
$$\begin{array}{ccccc} s & & p & & o \\ p & & \text{rdfs:domain} & & x \end{array} \implies s \quad & & \text{rdf:type} & & x$$
9.
$$\begin{array}{ccccc} s & & p & & o \\ p & & \text{rdfs:range} & & x \end{array} \implies o \quad & & \text{rdf:type} & & x$$

The `rdfs:range` of `rdfs:subClassOf` is `rdfs:Class`; apply rule #9



This is captured by RDFS inference rule #5

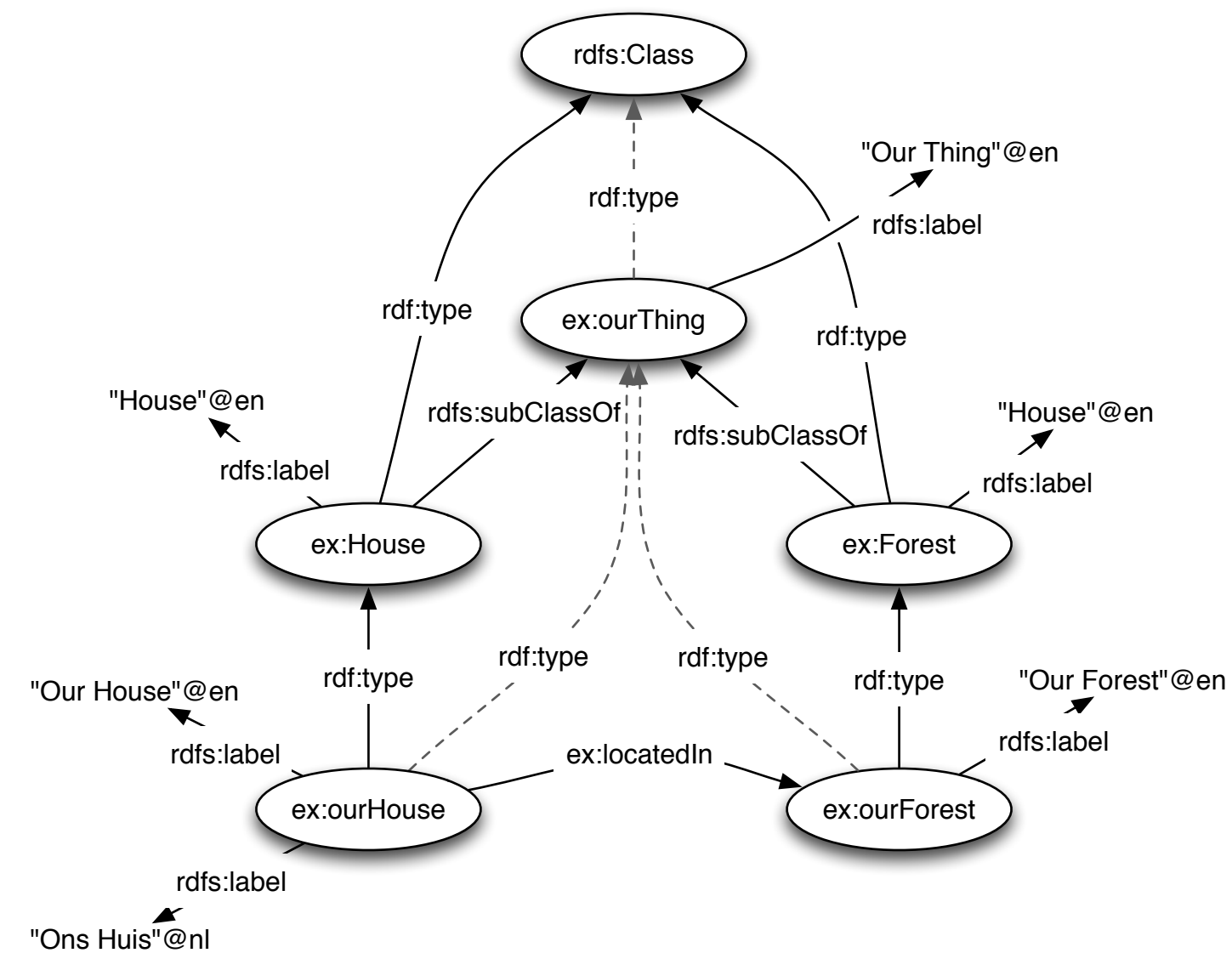
```

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4
5  ex:ourHouse rdf:type ex:House, ex:ourThing ;
6             ex:locatedIn ex:ourForest .
7
8  ex:House rdf:type rdfs:Class ;
9           rdfs:subClassOf ex:ourThing .
10 ex:Forest rdf:type rdfs:Class ;
11          rdfs:subClassOf ex:ourThing .
12
13 ex:ourForest rdf:type ex:Forest, ex:ourThing .
14
15 ex:ourThing rdf:type rdfs:Class .
16

```

5.
$$\begin{array}{ccccc} s & & \text{rdfs:subClassOf} & & y \\ y & & \text{rdfs:subClassOf} & & z \end{array} \implies s \quad \text{rdfs:subClassOf} \quad z$$
6.
$$\begin{array}{ccccc} p & & \text{rdfs:subPropertyOf} & & q \\ q & & \text{rdfs:subPropertyOf} & & r \end{array} \implies p \quad \text{rdfs:subPropertyOf} \quad r$$
7.
$$\begin{array}{ccccc} s & & p & & o \\ p & & \text{rdfs:subPropertyOf} & & q \end{array} \implies s \quad & & q & & o$$
8.
$$\begin{array}{ccccc} s & & p & & o \\ p & & \text{rdfs:domain} & & x \end{array} \implies s \quad & & \text{rdf:type} & & x$$
9.
$$\begin{array}{ccccc} s & & p & & o \\ p & & \text{rdfs:range} & & x \end{array} \implies o \quad & & \text{rdf:type} & & x$$

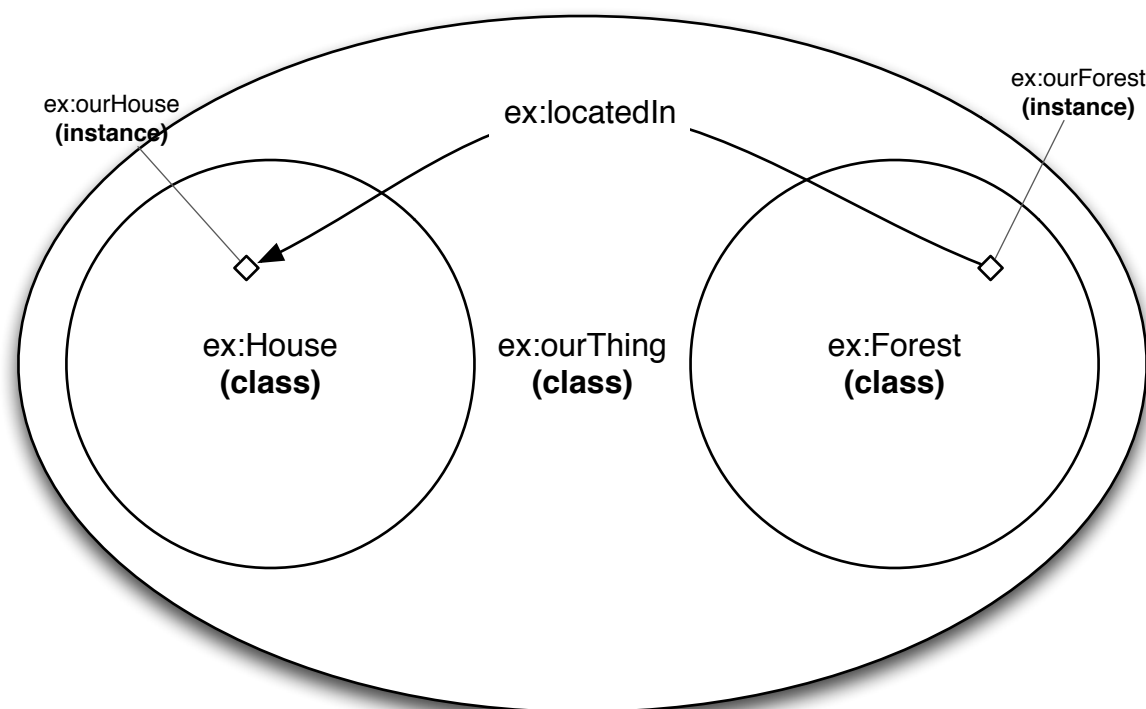


```

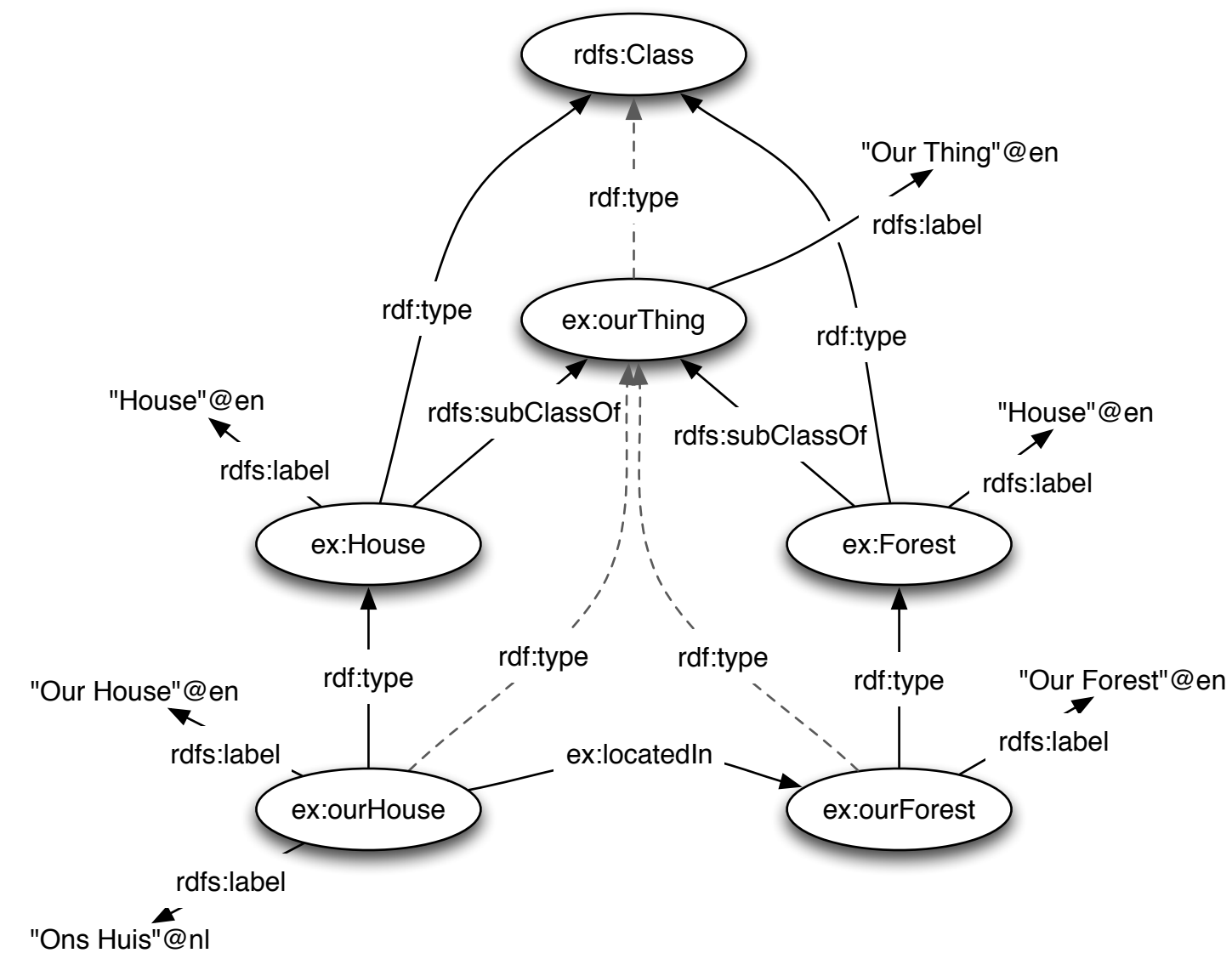
presentation-example.ttl — /Users/hoekstra/projects/semanticweb-rdf-...

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3  @prefix ex: <http://example.com/sw2016/> .
4
5  ex:ourHouse rdfs:type ex:House ;
6             ex:locatedIn ex:ourForest ;
7             rdfs:label "Our House"@en ;
8             rdfs:label "Ons Huis"@nl .
9
10 ex:House rdfs:type rdfs:Class ;
11          rdfs:subClassOf ex:ourThing ;
12          rdfs:label "House"@en .
13
14 ex:Forest rdfs:type rdfs:Class ;
15          rdfs:subClassOf ex:ourThing ;
16          rdfs:label "Forest"@en .
17
18 ex:ourForest rdfs:type ex:Forest ;
19             rdfs:label "Our Forest"@en .
20
21 ex:ourThing rdfs:type rdfs:Class ;
22            rdfs:label "Our Thing"@en .

```



Let's add some labels

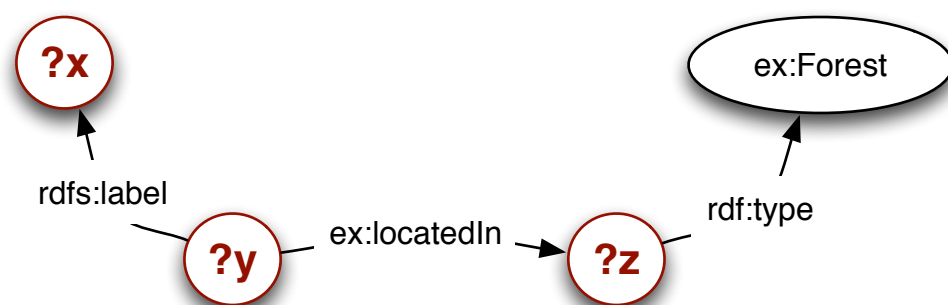


```

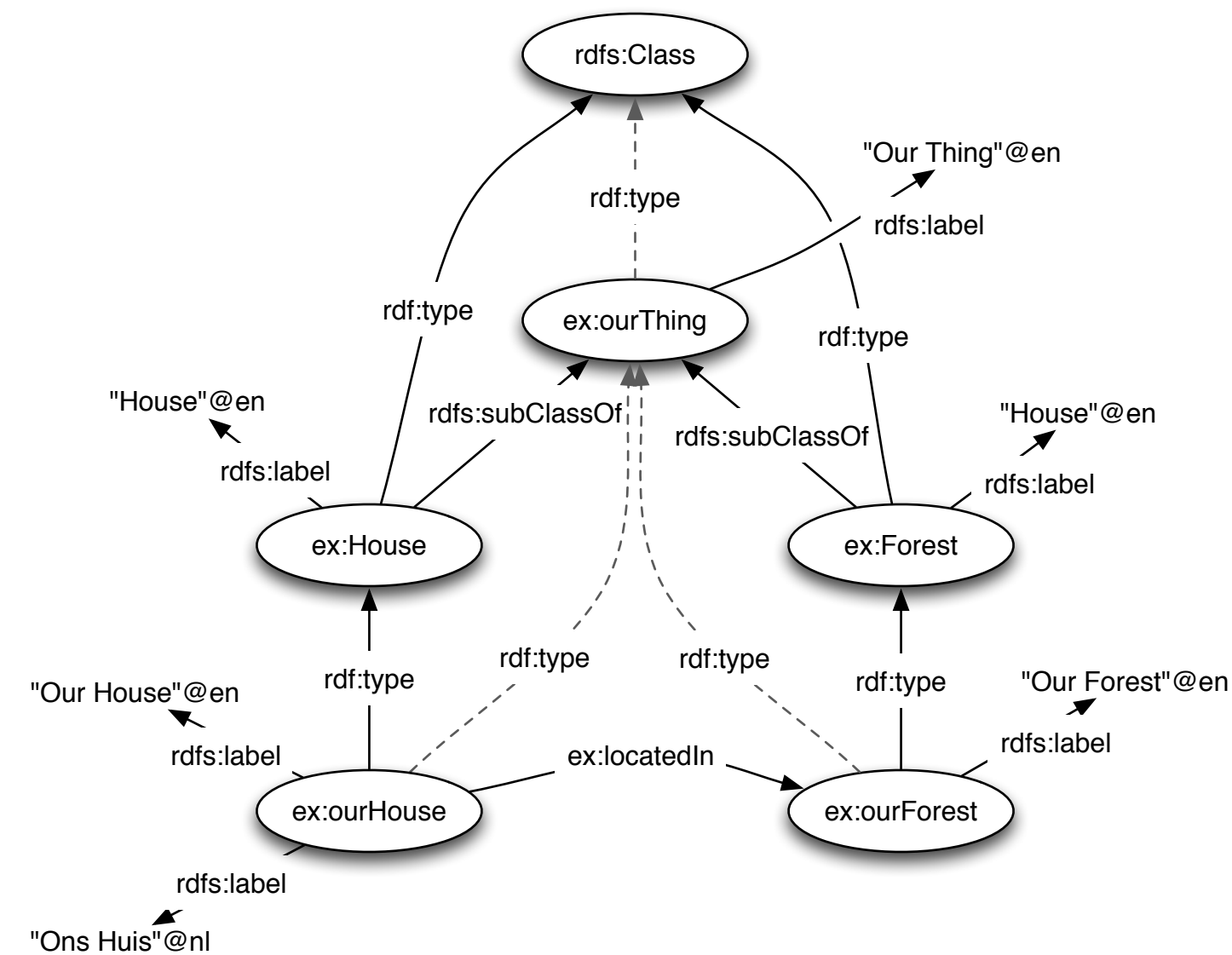
presentation-example.ttl — /Users/hoekstra/projects/semanticweb-rdf-...

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6             ex:locatedIn ex:ourForest ;
7             rdfs:label "Our House"@en ;
8             rdfs:label "Ons Huis"@nl .
9
10 ex:House rdfs:type rdfs:Class ;
11          rdfs:subClassOf ex:ourThing ;
12          rdfs:label "House"@en .
13 ex:Forest rdfs:type rdfs:Class ;
14          rdfs:subClassOf ex:ourThing ;
15          rdfs:label "Forest"@en .
16
17 ex:ourForest rdfs:type ex:Forest ;
18             rdfs:label "Our Forest"@en .
19
20 ex:ourThing rdfs:type rdfs:Class ;
21            rdfs:label "Our Thing"@en .
22
File 0 Project 0 ✓ No Issues presentation-example LF UTF-8 Turtle 7 updates

```



"All labels of all things that are located in a forest"

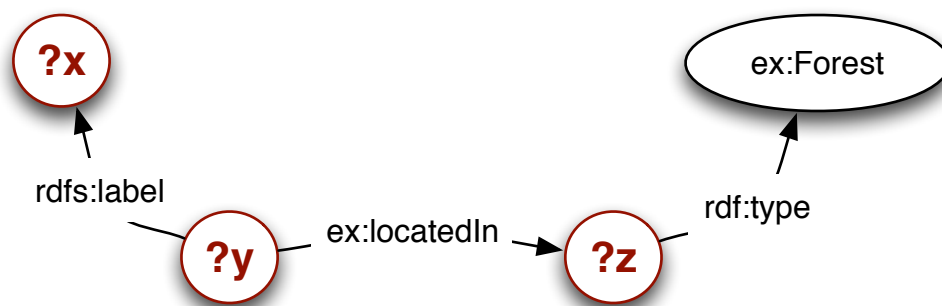


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5  ex:ourHouse rdfs:type ex:House ;
6             ex:locatedIn ex:ourForest ;
7             rdfs:label "Our House"@en ;
8             rdfs:label "Ons Huis"@nl .
9
10 ex:House rdfs:type rdfs:Class ;
11          rdfs:subClassOf ex:ourThing ;
12          rdfs:label "House"@en .
13 ex:Forest rdfs:type rdfs:Class ;
14           rdfs:subClassOf ex:ourThing ;
15           rdfs:label "Forest"@en .
16
17 ex:ourForest rdfs:type ex:Forest ;
18             rdfs:label "Our Forest"@en .
19
20 ex:ourThing rdfs:type rdfs:Class ;
21            rdfs:label "Our Thing"@en .
22
File 0 Project 0 ✓ No Issues presentation-example LF UTF-8 Turtle 7 updates

```



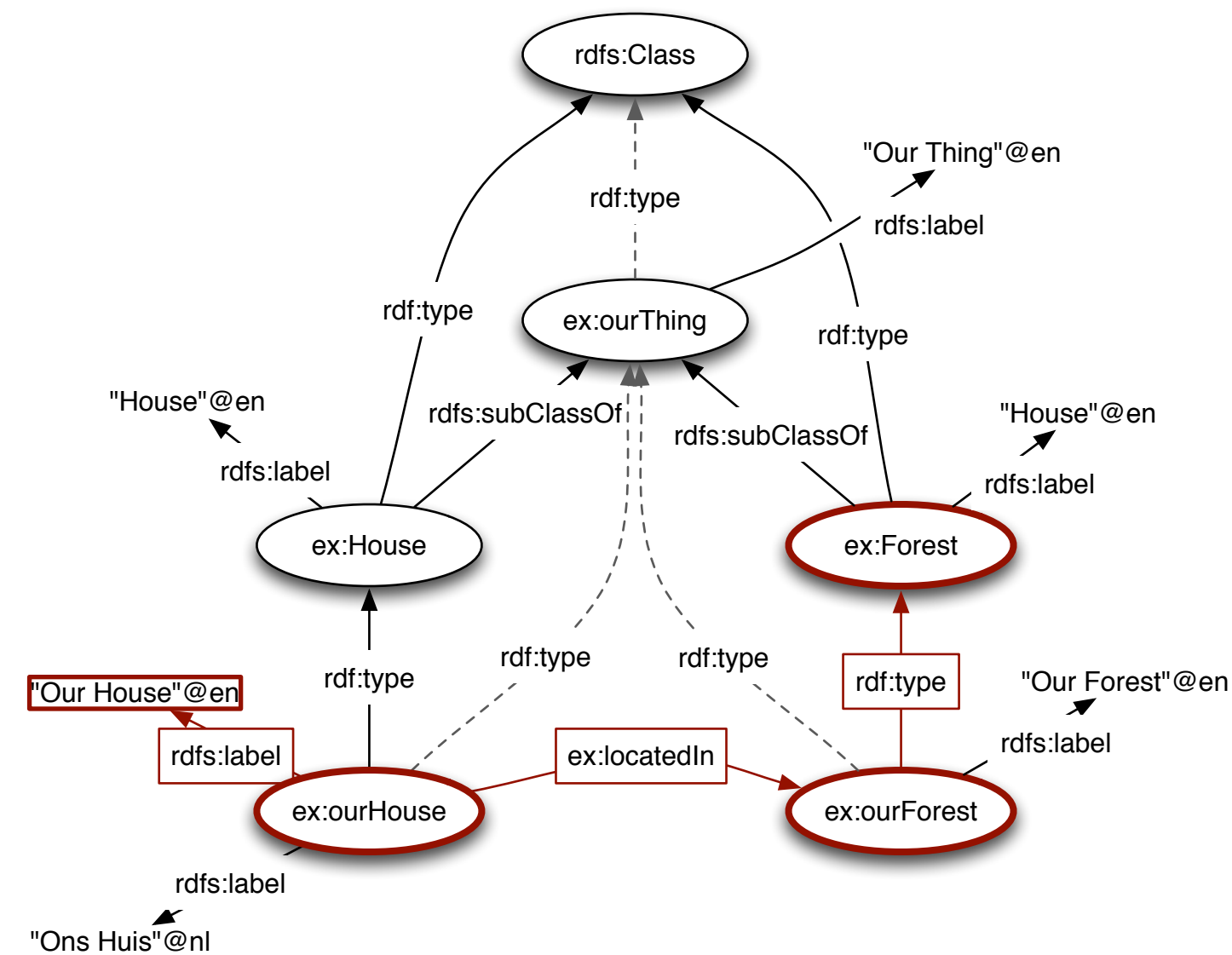
```

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX ex: <http://example.com/sw2016/>

SELECT ?x WHERE {
  ?y rdfs:label ?x .
  ?x ex:locatedIn ?z .
  ?z rdf:type ex:Forest .
}

```

The SPARQL query is just a graph pattern that we're trying to 'fit'

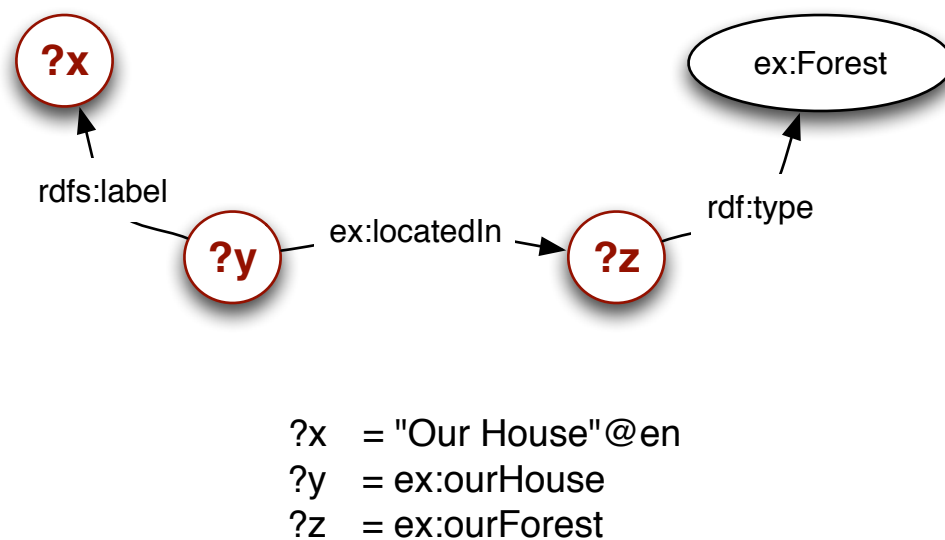


```

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1  @prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
2  @prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
3  @prefix ex: <http://example.com/sw2016/> .
4
5  ex:ourHouse rdf:type ex:House ;
6             ex:locatedIn ex:ourForest ;
7             rdfs:label "Our House"@en ;
8             rdfs:label "Ons Huis"@nl .
9
10 ex:House rdf:type rdfs:Class ;
11          rdfs:subClassOf ex:ourThing ;
12          rdfs:label "House"@en .
13 ex:Forest rdf:type rdfs:Class ;
14           rdfs:subClassOf ex:ourThing ;
15           rdfs:label "Forest"@en .
16
17 ex:ourForest rdf:type ex:Forest ;
18             rdfs:label "Our Forest"@en .
19
20 ex:ourThing rdf:type rdfs:Class ;
21            rdfs:label "Our Thing"@en .
22
File 0 Project 0 ✓ No Issues presentation-example LF UTF-8 Turtle 7 updates

```



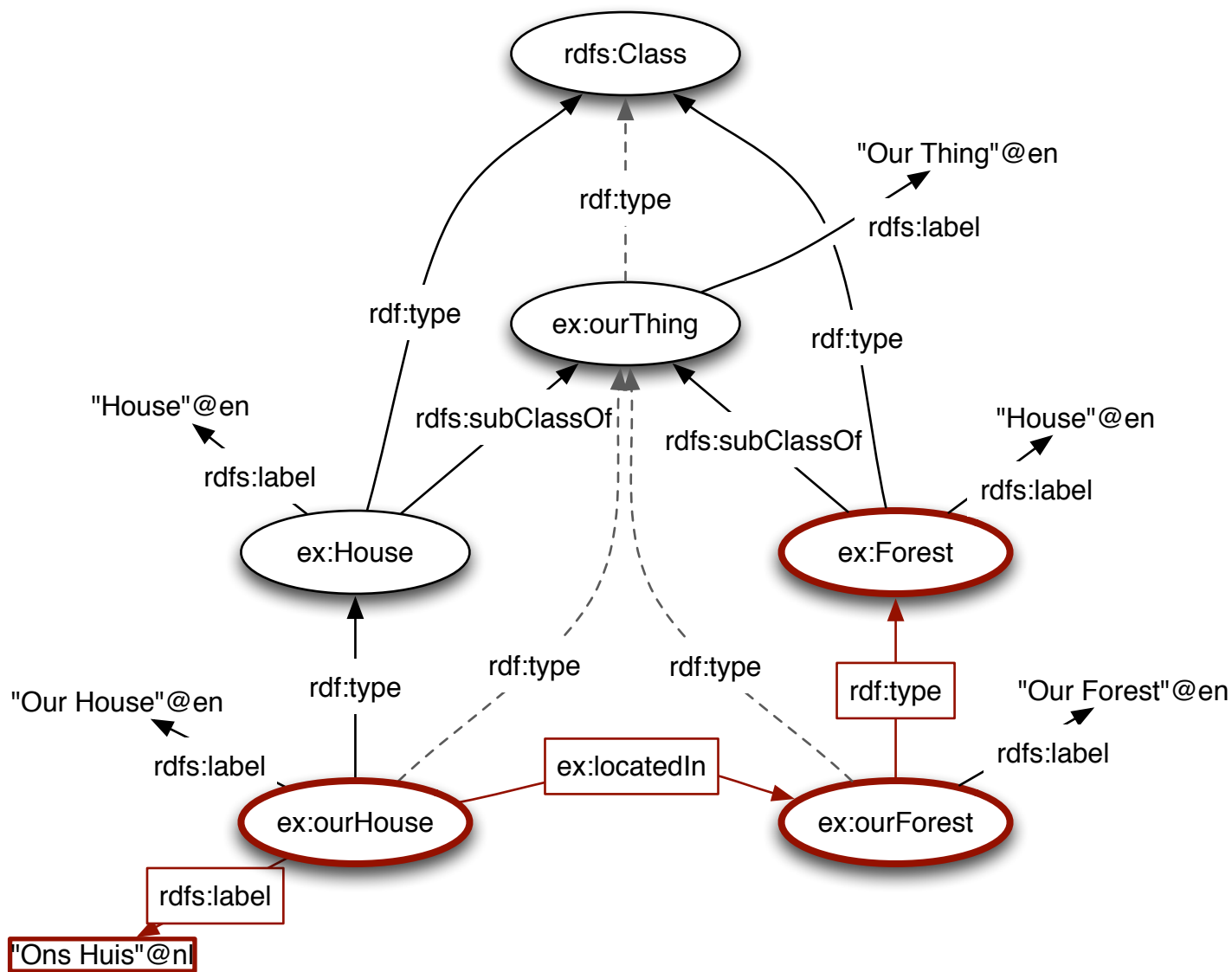
It matches a part of our graph!

```

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX ex: <http://example.com/sw2016/>

SELECT ?x WHERE {
  ?y rdfs:label ?x .
  ?x ex:locatedIn ?z .
  ?z rdf:type ex:Forest .
}

```

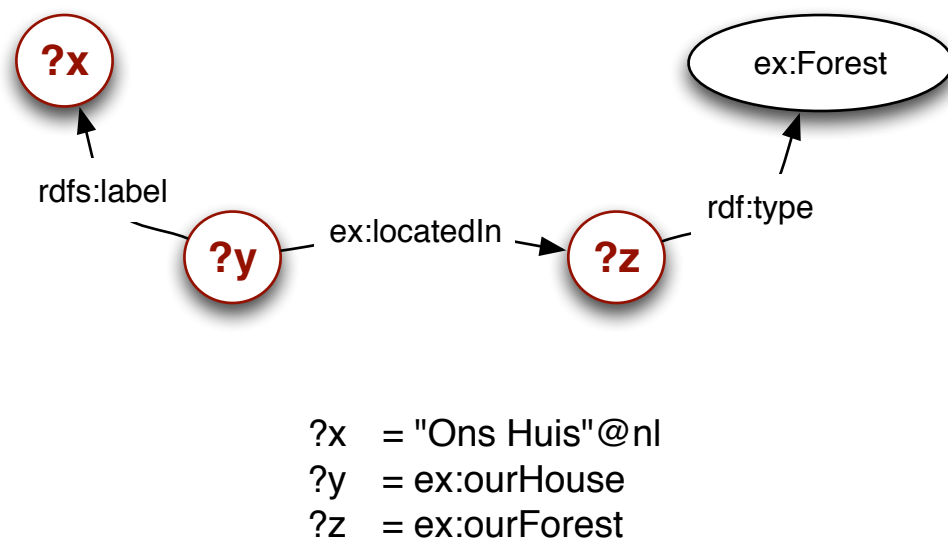


```

presentation-example.ttl — /Users/hoekstra/projects/semanticweb-rdf-...

presentation-example.ttl
1  @prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
2  @prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
3  @prefix ex: <http://example.com/sw2016/> .
4
5  ex:ourHouse rdf:type ex:House ;
6             ex:locatedIn ex:ourForest ;
7             rdfs:label "Our House"@en ;
8             rdfs:label "Ons Huis"@nl .
9
10 ex:House rdf:type rdfs:Class ;
11          rdfs:subClassOf ex:ourThing ;
12          rdfs:label "House"@en .
13 ex:Forest rdf:type rdfs:Class ;
14           rdfs:subClassOf ex:ourThing ;
15           rdfs:label "Forest"@en .
16
17 ex:ourForest rdf:type ex:Forest ;
18             rdfs:label "Our Forest"@en .
19
20 ex:ourThing rdf:type rdfs:Class ;
21            rdfs:label "Our Thing"@en .
22
File 0 Project 0 ✓ No Issues presentation-example LF UTF-8 Turtle 7 updates

```



"All labels of all things that are located in a forest"

```

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX ex: <http://example.com/sw2016/>

SELECT ?x WHERE {
  ?y rdfs:label ?x .
  ?x ex:locatedIn ?z .
  ?z rdf:type ex:Forest .
}

```